

INTRODUCTION TO DATA SCIENCE (IDS)

QUESTION PAPER – SET 1

Time: 3 Hours | Max Marks: 60

PART – A

Answer ALL Questions ($10 \times 1 = 10$ Marks)

1. Define Data Science.
2. What is Data Analytics?
3. Define Normal Distribution.
4. What is skewness?
5. Define missing data.
6. What is feature selection?
7. Define outlier.
8. What is Matplotlib?
9. Define supervised learning.
10. What is reinforcement learning?

PART – B

11.
 - a) Explain Data Science and its applications in detail. (5M)
 - b) Describe the Data Analytics Life Cycle with neat diagram. (5M)

OR

- c) Explain various tools used in Data Science Toolkit. (5M)
 - d) Differentiate between Normal Distribution and Skewed Distribution with examples. (5M)

12.

- a) Explain Normal Distribution with suitable graph and properties. (5M)
 - b) Solve problems related to mean, variance, and standard deviation. (5M)

OR

- c) Explain skewed distribution and its types with diagrams. (5M)
 - d) Write short notes on probability distribution formulas used in Data Science. (5M)

13.

- a) Explain different methods of handling missing data. (5M)
 - b) Discuss Feature Selection techniques with suitable examples. (5M)

OR

- c) What are outliers? Explain methods to detect and remove outliers. (5M)

d) Explain preprocessing steps in Data Science. (5M)

14.

a) Explain Data Reduction techniques in detail. (5M)

b) Discuss dimensionality reduction and sampling methods. (5M)

OR

c) What is Matplotlib? Explain advantages and disadvantages. (5M)

d) Write syntax and purpose of different plots in Seaborn and Matplotlib. (5M)

15.

a) Define Machine Learning and explain its applications. (5M)

b) Explain types of Machine Learning. (5M)

OR

c) Differentiate between Supervised and Unsupervised Learning. (5M)

d) Explain Reinforcement Learning with suitable example. (5M)